

Educonnect: A Peer-to-Peer and Collaborative Learning System for Students

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Declaration and Approval

We declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of our knowledge and belief, the research proposal contains no material previously published or written by another person except where due reference is made in the research proposal itself.

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Abstract

Peer support amongst students at university levels becomes increasingly difficult due to the limited availability of lecturers, large student numbers, and lack of real-time collaboration tools. Current solutions such as Piazza, Brainly, and Discord have attempted to solve certain issues, yet none have provided an all-encompassing platform combining the following features: real-time collaboration, automated study groups with video call integrations, gamified engagement, and a repository of community resources. This paper discusses the proposal of EduConnect, a website that provides a solution to these challenges by enabling students to interact effectively on the platform. Key features include the forum-like problem solving tool, live chat powered by WebSockets, automatic study group creation linked to Google Meet and Zoom API, gamification engine based on user streaks, and a resource repository ranked by the community. The application will be coded using React.js for the front-end, Django for the back-end, and PostgreSQL as the database. In addition, the website will be built on a JWT-authentication mechanism. The proposed method of development will involve using Agile Scrum and OOAD methodologies throughout the requirement analysis, system design, implementation, testing, and deployment stages.

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Abbreviations

API	Application Programming Interface
DRF	Django REST Framework
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICS	Informatics and Computer Science
IR	Information Retrieval
JSON	JavaScript Object Notation
JWT	JSON Web Tokens
LMS	Learning Management System
OOAD	Object-Oriented Analysis and Design
ORM	Object-Relational Mapper
OAuth	Open Authorization
Q&A	Question and Answer
REST	Representational State Transfer
SCES	School of Computing and Engineering Sciences
TCP	Transmission Control Protocol
UI	User Interface
URL	Uniform Resource Locator

Chapter 1: Introduction

1.1 Background Information

The education sector has been significantly transformed due to the rapid and aggressive growth of technology, particularly in how students can access knowledge for research and how they can collaborate with other students. Education is swiftly shifting from the olden and traditional classroom-based interactions and moving on to more digital and interactive learning platforms that can enable easier collaboration among students, better learning resources and engagement. That is why technological platforms that help facilitate student interaction, such as discussion forums and collaborative tools have become essential in today's world in helping tackle academic challenges (Bond et al.,2020). Despite all these advancements students will still struggle in being able to find support from outside the normal teaching environments like classrooms. This study hence focuses on the development of a web-based platform, EduConnect, aimed at enhancing peer-to-peer academic support through collaborative problem-solving and resource sharing.

The issue of having very little access to immediate academic assistance among students, especially those in higher education institutions, remains widely occurring. Many students have and will continue to experience delays in receiving help due to how limited lecturers are, large number of students, and lack of an organized peer support system. Learning is significantly influenced through social interaction and collaborations (Siemens, 2018). Our system today hence fails to effectively address these interactions. lack of real-time engagement in an online learning environment reduces how effective the lessons are, and the participation of the students becomes very low which may result in increased dropout rates and reduced quality of learning outcomes (UNESCO, 2021). This hence confirms that the lack of interactive platforms is a problem worth looking at.

This problem affects a very large number of people including students, educational institutions and educators. Students find it difficult to understand challenging concepts, which leads to poor academic performance. Educational institutions get low student satisfaction and morale, whereas the instructors are burdened with ever repeating questions which would be easily solved through peer collaboration. If the issue is not addressed the number of dropouts will significantly increase and get a reduced quality of learning outcomes.

Researchers and academicians have tried to solve this problem through online forums, learning management systems and collaborative tools among various other solutions. Some

have worked such as discussion boards and Q&A systems in promoting knowledge sharing, however, they often lack real-time interaction, automation and engagement mechanisms such as gamification (Khalil & Ebner, 2020). These recent developments have henceforth shown the importance of integration of real-time communication, motivational features to enhance user engagement. Despite all of these, there is still a huge gap left in creating a localized, student-driven platform that can bring forward collaboration, automation and gamification all in one. This study brings forward EduConnect as a solution to bridge these gaps by providing a dynamic, interactive and student-centred environment.

1.2 Problem Statement

Students in universities and higher educational institutions have a hard time getting effective support outside of the classroom setting. This particularly affects students who may require clarification on certain difficult topics but may have limited access to their lecturers or lack a structured peer support system. Citing Martin and Bolliger (2018), Insufficient interaction in online learning environments leads to reduced engagement and ineffective learning outcomes.

The problem stems from delayed problem-solving, limited collaboration and inefficient knowledge sharing among students. It came to realization that most of the platforms in use today do not support real-time communication, automated collaboration and gamified experience which help one's engagement. Hence students end up taking a lot of their time looking for adequate learning resources and some of them even end up not looking at all. Studies show that collaborative learning significantly improves understanding, but only when supported by effective technological tools (Bond et al.,2020).

This is a problem since the ineffective academic platforms may lead to poor academic performance and a huge dip in the morale of the students. In addition, lecturers get tasked with getting asked very repetitive questions. If the gap remains unaddressed the learning experience of many students may end up being very unpleasant and very hindered which will limit their academic development. UNESCO (2021) puts emphasis on the fact that the lack of academic integration is a major factor contributing to student attrition, highlighting the importance of addressing the issue through innovative solutions.

1.3 Objectives of the study

To develop a web-based collaborative platform which enhances peer-to-peer academic support by enabling real-time problem solving, automated study group formation and resource sharing among students.

1.3.1 Specific Objectives

- I. To investigate the current problems that students are facing in getting peer-based academic support within learning institutions.
- II. To analyse and identify the gaps in the existing solutions.
- III. To design a web-based system architecture for EduConnect.
- IV. To develop EduConnect with real-time communication and automated group formation features.
- V. To test and evaluate the functionality and performance of the developed EduConnect system.

1.3.2 Research Questions

- I. What challenges do students face in accessing peer-to-peer academic support?
- II. What are the problems faced by the current academic platforms?
- III. How can a web-based system be designed to support real-time academic collaboration?
- IV. How effective is the developed EduConnect in enhancing collaboration and problem solving?
- V. How easy is the usage and reliability of the EduConnect system to students, lecturers and educational institutions?

1.4 Justification

This study is essential in addressing the growing need for efficient, collaborative and easily accessible educational systems. Through EduConnect, the research will be able to help students improve their learning experiences and improve their academic performance through the real-time collaboration feature and peer support system. The students will be able to get quick and readily available assistance, develop study groups and access academic resources easily. Through this the students will be able to improve their academic performance, which will therefore benefit the educational institutions through engagement and satisfaction. The

instructors will get a load off their shoulders as students will become more self-reliant hence carrying less of a burden on their shoulders.

Looking at it from a different angle, the research supports the developing educational and learning technologies which aligns with the growing global trend of education through technology. Governments and educational stakeholders can use such systems to their advantage to improve the quality of education. Literature supports the integration of collaborative technologies to enhance learning outcomes and student retention (Bond et al.,2020), reinforcing the relevance of this study.

1.5 Scope

The proposed study focuses on the development of EduConnect, a localized, student-driven web platform designed to facilitate peer-to-peer academic collaboration. EduConnect will involve creating a forum-style interface for academic problem-solving, an automated study group formation system using the Google Meet/Zoom API, and a gamified engagement system based on "streaks" to reward active participation. The study will target students and "expert solvers" within the higher education community, primarily focusing on the web-based environment using React.js for the front end and Django for the backend. Boundaries of the project include the use of PostgreSQL for database management and JWT for secure user authentication. The project will not include the development of a native mobile application, nor will it incorporate complex learning management system (LMS) features like grading or course enrolment, as the intent is to remain a student-centred support tool rather than a formal instructional environment.

1.6 Limitations

Challenges anticipated during the development of this platform include the technical complexity of integrating third-party APIs for automated meeting generation and ensuring real-time communication stability through WebSockets. Additionally, the study may face difficulties in recruiting enough "expert solvers" to provide initial content and solutions on the platform, which is critical for establishing a viable knowledge base. There is also the potential constraint of limited student participation during the initial testing phase, which may hinder the evaluation of the automated group formation features.

1.7 Delimitations

To tackle these anticipated challenges, EduConnect will implement a robust testing phase to ensure the reliability of WebSockets and API integrations before deployment. To address the scarcity of initial content, a "streak" and reward system will be prioritized early in development to incentivize early adopters and solvers to engage with the platform. Furthermore, the study will focus on a specific localized student community during the evaluation phase to ensure a high number of students, thereby facilitating more effective testing of the automated collaboration and group-matching algorithms.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the existing literature related to peer-to-peer academic support systems, collaborative learning technologies and digital platforms used in education. It dives into the current processes that are in use to facilitate academic collaboration, identifying challenges in existing systems and looking at the solutions which were found in the past. This chapter further highlights the gaps which are in the current approaches and discusses technologies relevant to the proposed EduConnect system. The conceptual framework will illustrate how the proposed solution will address the gaps found.

2.2 Existing technologies curbing peer-to-peer and collaborative writing

The education sector has been adopting technology recently to improve learning and collaboration among students. Systems such as Learning Management Systems, discussion forums and online collaboration tools have been used to facilitate communication and knowledge. The systems support processes such as assignment submission, discussion-based learning and resource sharing among students and instructors. Collaborative learning, which helps promote interaction among learners, has been identified as a key factor in improving understanding and engagement (Otto et al., 2023)

This study will mainly focus on peer-to-peer academic support where the students will be able to assist each other in understanding certain concepts and solving problems. This is currently taking place on message platforms, informal study groups and online forums. As useful as these methods have proven to be they often lack structure and real-time responsiveness. Many platforms do not provide automated mechanisms for matching students or facilitating immediate interaction which result in delays in problem-solving (Zhang et al., 2024).

The academic platforms which have proven to be effective are guided by accessibility, timely feedback and collaboration as the major principles. Educational theories emphasize the importance of interaction and engagement in learning environments (Priyatno et al., 2024).

These principles are often traded off due to the limited system capabilities, inadequate engagement features and lack of real-time communication. The expectation is a seamless, interactive learning environment but it is fragmented systems which fail to adequately support the students through collaborative learning which leaves a gaping hole this study looks to address.

2.3 Challenges Facing Peer-to-peer and collaborative learning

2.3.1 Lack of real-time interaction

The lack of real-time interaction is of the major challenges ahead in peer-to-peer academic support systems. As many platforms tend to rely on asynchronous communication, delayed responses and reduced engagement is usually a major setback. Additionally, there is a lack of structured collaboration which makes it hard for students to find appropriate peers or resources. (Zhang et al., 2024).

2.3.2 Low engagement

Low user engagement is another challenge as the existing will often fail to motivate the students to actively participate due to the absence of engagement mechanisms such as gamification (Moon et al., 2024). They will reduce the effectiveness and usage of the platforms

2.3.3 Poor User Experience

Some systems have been found to be unscalable and inaccessible. Large student populations make it difficult to provide personalized support while some platforms are not optimized for easy use. If the challenges were to remain unattended too, they may lead to poor academic performance, reduced student satisfaction and lastly increased number of dropouts. (Moon et al., 2024).

2.3.4 Lack of Real-time Communication

LMS platforms and discussion forums are a few of the solutions to have emerged, but they still lack integration of real-time communication, automation and engagement features. This leaves huge gaps that must be addressed.

2.4 Review of Existing Systems

Several systems have been developed to address the challenge of peer-to-peer academic support and collaborative learning among students in higher education. The Systems include Piazza, Brainly and Discord. This section will review their purpose, functionality, underlying technologies and Limitations and the gap that Educonnect seeks to fill.

2.4.1 Piazza

Piazza is a cloud-based question and answer based platform founded in 2009, it was originally designed to address communication barriers experienced in large university lecture halls. The platforms operate as a wiki-style forum that enables students, teaching assistants and instructors to collaborate, ask and answer questions and share course related resources. Piazza functions as a class specific discussion board where instructors enrol students by emails or sign-up links. Once enrolled, students can post questions publicly or privately and both students and instructors can collaboratively edit answers in a wiki-style format. Instructors can endorse correct answers, prioritize unanswered questions and use a polling feature to gather feedback. The platform supports integration with major Learning Management Systems (LMS) including Blackboard, Moodle, Sakai, Canvas and Coursera making it straightforward to embed within existing institutional infrastructure. Piazza also supports anonymous posting. Despite these strengths, Piazza presents notable limitations relative to the problem EduConnect aims to solve. The platform is primarily asynchronous and does not support real-time live chat or instant messaging between students, which limits its usefulness for urgent problem-solving. The platforms open visibility also raises privacy concerns, students may feel discouraged from asking questions where peers can also see their posts(SoftwareAdvice,2026). Piazza has further been criticised for information overload, as high volumes of notifications generated in large classes can become overwhelming and reduce meaningful engagement. Critically, Piazza offers no gamification or engagement incentives, no automated study group formation and no integration with video conferencing tools. Students are therefore unable to transition seamlessly from asynchronous text-based discussion into real-time collaborative sessions

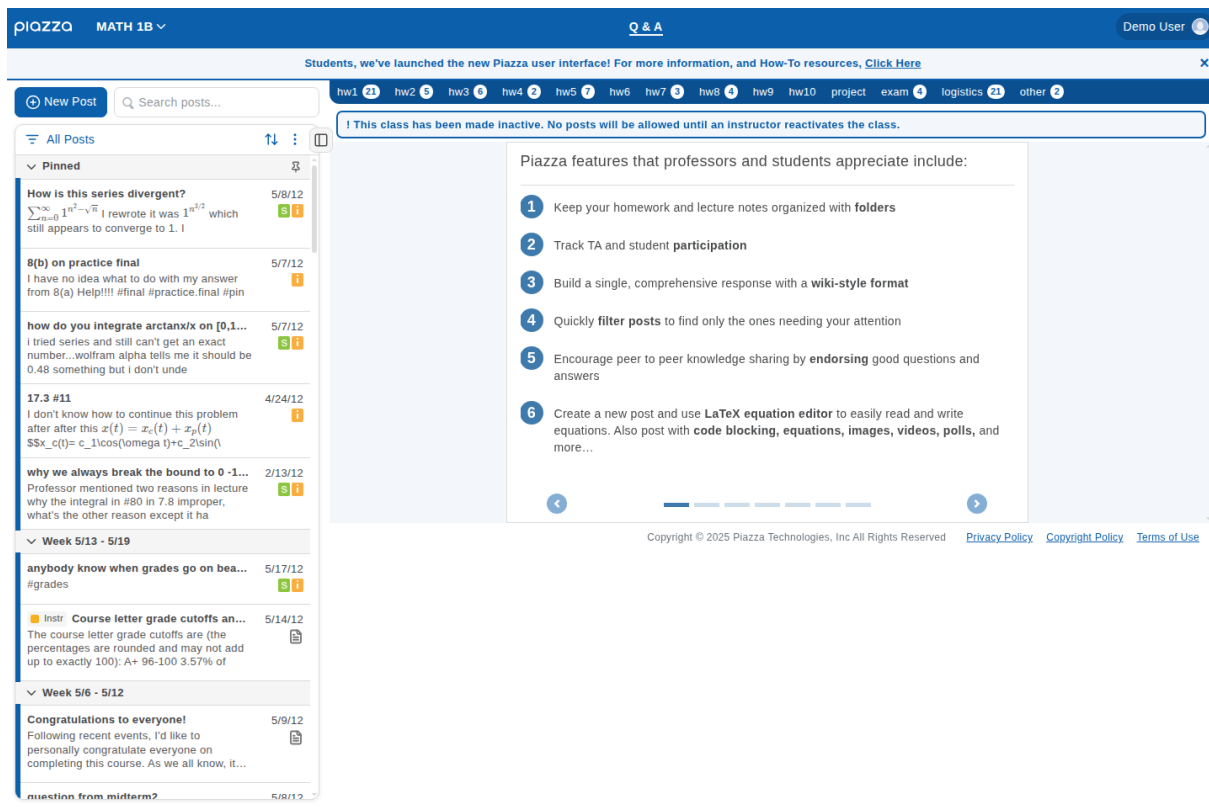


Figure 1 ,Piazza Q&A Forum Interface (Piazza,2026)

2.4.2 Brainly

Brainly is crowd-sourced homework help and community learning platform founded in 2009. Brainly's stated mission is to provide equal access to high quality educational support enabling any student, regardless of financial background to access help on demand.

The platform operates as a question-and-answer community where students post academic questions and receive responses from peers, educators or subject matter experts. Questions are organised by subject, and users earn points for providing correct and helpful answers a basic gamification mechanism intended to foster community participation (SoftwareWorld, 2026). Brainly integrated artificial intelligence features powered by OpenAI GPT-4 adding an AI Tutor that provides step-by-step explanations, personalised study plans and AI-generated practice questions. Brainly faces several critical gaps that EduConnect addresses. The quality of community-submitted answers is highly variable users consistently note that answers can be short, poorly explained or in some cases incorrect creating a risk that students reinforce

incorrect understanding (Capterra, 2026). The platform also lacks real-time communication capabilities; there is no live chat, instant messaging or video conferencing. Study group formation is entirely manual with no automated mechanism to match students based on shared academic challenges or availability. Furthermore, while Brainly's gamification is limited to a basic points and ranking system, it does not include streak-based daily engagement incentives or collaborative reward structures. These shortcomings highlight a clear opportunity for EduConnect to deliver a more structured, localised and real-time collaborative environment.

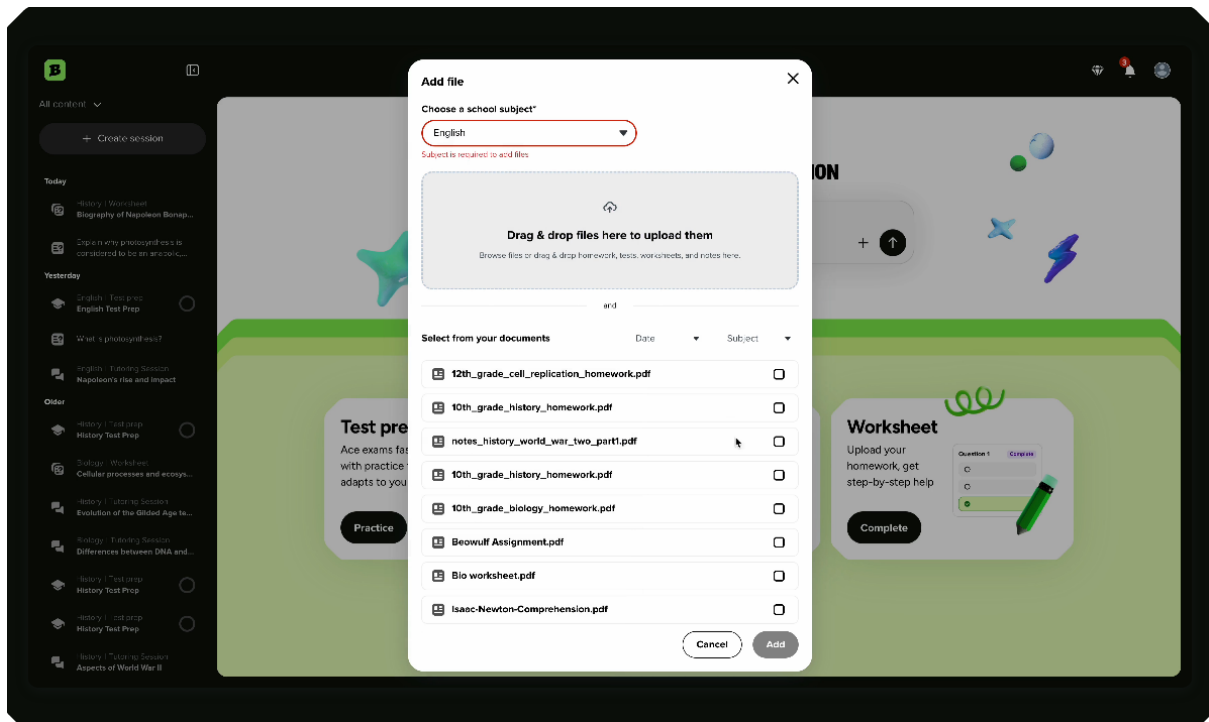


Figure 2, Brainly Question-and-Answer Platform Interface (MyEngineeringBuddy, 2026)

2.4.3 Discord (Academic Study Servers)

Discord is a free communication platform initially launched in 2015 for gaming communities which has since evolved into a versatile tool used across diverse communities including academic study groups. Discord's adaptability and zero-cost access model have made it popular among university students who use it to organise study groups, share resources, and collaborate on academic tasks both inside and outside formal course structures. Discord operates through a server-based model where users create or join communities organised around shared interests, subjects or projects. Each server comprises text channels for written discussion and voice channels for audio or video communication. Key features relevant to

academic use include real-time text and voice communication, screensharing for collaborative problem-solving, customisable roles and permissions and integration of bots for scheduling, reminders, task management and interactive quizzes (ThinkAcademy, 2025). Notwithstanding these benefits, Discord presents significant limitations when considered as a dedicated academic support platform. Its origins as a gaming tool mean that it lacks purpose-built academic features there is no structured question and answer system, no academic resource repository, no expert-solver role management and no automated study group matching. Discord also lacks gamification or streak-based engagement mechanisms, has no integration with meeting tools for automated session scheduling and provides no notification system tied to academic workflows. These gaps confirm that while Discord enables real-time communication, it requires substantial customisation and lacks the academic-specific structure that EduConnect is designed to provide.

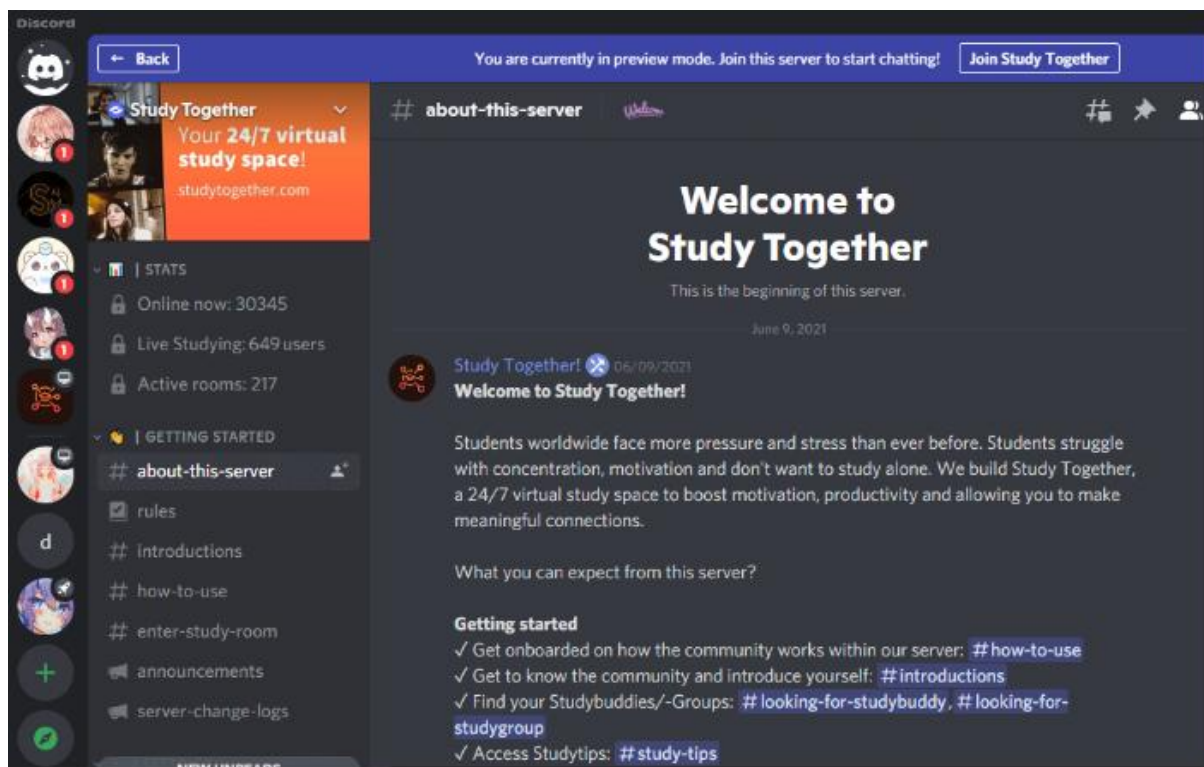


Figure 3, Discord Academic Study Server Interface (Evepaper, 2025)

2.5 Gaps in Current Approaches and Related Applications

The review of existing systems reveals a consistent pattern of partial solution: each of the three platforms addresses some dimension of peer academic support, yet none of them provides a unified, localised platform that combines real time communication, automated

study group formation, gamified engagement and structured resource sharing. The following gaps are identified across the reviewed systems and directly inform the design basis for Educonnect. The first and most common gap is the absence of real time, synchronous communication within purpose build academic platforms. Piazza and Brainly operate predominantly on asynchronous models, students post questions and must wait for responses that can range from minutes to days depending on community activity. While Discord supports real time voice and text it lacks the academic structure needed to make such communication educationally productive. Educonnect address this gap through WebSocket powered live chat and real time notifications, enabling students to engage in instant collaborative problem solving.

The second significant gap is the lack of automated peer matching and study group formation. In the reviewed systems, students must manually search for peers with similar academic needs, which is both inefficient and often unsuccessful in large, unstructured communities. None of the three platforms includes an algorithm or automation mechanism that groups students by shared subject interest, academic difficulty, or schedule availability. EduConnect's automated study group formation feature which leverages shared topic tags and integrates with Google Meet or Zoom to generate unique meeting links directly addresses this deficiency.

The third gap concerns the inadequacy of gamification and engagement mechanisms. While Brainly employs a basic points system and Discord provides server roles, neither offers sustained daily engagement features such as activity streaks, tiered achievements or collaborative reward structures. EduConnect's streak system and reward-based participation model are designed to sustain student motivation beyond initial platform sign-up.

The fourth gap is the absence of a dedicated resource repository within collaborative platforms. While Brainly organises questions by subject, it does not provide a structured space where students can recommend, and rank learning resources such as textbooks, articles or study websites tied to specific topics. Piazza allows file sharing but only within the confines of instructor moderated class threads. EduConnect fills this gap with a standalone resource repository where students can submit rate and filter academic materials independently of Q&A threads.

2.6 Technologies Related to the Study

This section reviews the key technologies that underpin EduConnect's proposed solution.

2.6.1 WebSockets

WebSockets provide a full-duplex, bidirectional communication channel over a single, persistent TCP connection. Unlike standard HTTP, which follows a rigid request-response model and closes after each interaction, WebSockets keep a continuous link open between the client and server. This allows for instant data exchange from either side without the lag of repeated handshakes (LogRocket, 2025). Research shows that WebSockets outperform HTTP long-polling by reducing average latency by roughly 70%—dropping from 100ms to just 30ms while cutting bandwidth usage in half. This makes them the superior choice for real-time collaboration (IJNRD, 2024).

In EduConnect, WebSockets will drive the live chat in study groups, Q&A threads, and the real-time notification system for alerts like new answers or meeting reminders. To manage this, we will use Django Channels, an extension that handles these connections asynchronously. This ensures the backend can support numerous student interactions simultaneously without slowing down (Honeybadger, 2026). This proven approach aligns with EduConnect’s mission to provide a seamless, interactive environment for peer support.

2.6.2 RESTful API Architecture

Representational State Transfer (REST) is an architectural style designed for networking applications via stateless HTTP requests. It uses clearly defined endpoints to create, read, update, or delete data. For EduConnect, the Django REST Framework (DRF) serves as the bridge between the React.js frontend and the Django backend for standard tasks like managing profiles, creating forum threads, and scheduling groups (DigitalOcean, 2025). By using REST for data management and WebSockets for real-time events, the system maintains a "separation of concerns," ensuring the architecture remains clean and scalable.

2.6.3 JSON Web Tokens (JWT)

JSON Web Tokens are a secure, compact way to transmit information as a JSON object. When a user logs into EduConnect, the backend issues an access token and a refresh token. The client includes this access token in the header of future requests to verify identity. Because these tokens are stateless and carry all necessary user data, the server does not need to store active sessions, which significantly boosts scalability. We will use the Simple JWT library to manage this, allowing us to easily distinguish between regular students and "Expert Solvers" who have special posting privileges (DigitalOcean, 2025).

2.6.4 Video Conferencing API Integration

To make collaboration seamless, EduConnect automates meeting generation. Instead of forcing students to manually create and share links, the platform uses the Google Meet or Zoom API to generate unique links automatically whenever a study group forms. This allows students to jump from text chat to live video without any friction. These APIs use OAuth 2.0 for security and are triggered directly by backend logic (DigitalOcean, 2025). This level of automation is a standout feature, setting EduConnect apart from the existing systems reviewed in Section 2.4.

2.7 Conceptual Framework

EduConnect is conceived as a localised, web-based, student-driven academic collaboration platform that addresses the identified gaps in peer-to-peer academic support. The conceptual framework below describes how the proposed solution is intended to operate from data input through processing and storage to output and reporting for each category of stakeholder involved in the system. The framework is organised around four interconnected components: input, processing, storage, and output, and recognises two primary user roles: Students (general users) and Expert Solvers (verified advanced students or teaching assistants with elevated contribution privileges).

With respect to input, students interact with EduConnect by posting academic problems or questions through a structured forum-style interface, submitting recommended study resources such as textbooks and URLs, registering for accounts and selecting subject areas of interest. Expert Solvers provide answers, edits and verifications. The input layer is mediated by the React.js frontend, which provides a responsive, component-based user interface for all data entry and interaction.

With respect to processing, the Django backend receives all inputs through RESTful API endpoints and WebSocket consumers. Core processing operations include: authenticating and authorising users via JWT tokens with role-based access control; routing forum questions to relevant subject categories and notifying matched Expert Solvers in real time via WebSockets; automatically forming study groups by matching students who have expressed interest in the same subject areas or posted similar questions, and calling the Google Meet or Zoom API to generate a unique meeting link for each group; tracking daily participation events such as posting, answering, and attending session and computing streak scores to

update each user's gamification status; and processing resource submissions and ranking them based on community upvotes.

With respect to storage, all data persisted in a PostgreSQL relational database, chosen for its robustness, support for complex queries and compatibility with Django's Object-Relational Mapper (ORM). The database schema stores user profiles, forum threads and responses, resource entries, study group records, meeting links, streak and points data, and notification logs.

With respect to output, the system delivers value to students through: a categorised, searchable feed of academic questions and expert-verified answers; real-time notifications for new responses, study group formations and upcoming meeting reminders; automatically generated meeting links embedded within study group pages; a personal dashboard displaying current streaks, accumulated points, and contribution history and a ranked resource repository organised by subject and community rating. Institutional administrators can access usage analytics dashboards showing platform activity trends, which supports evidence-based decisions about student support interventions.

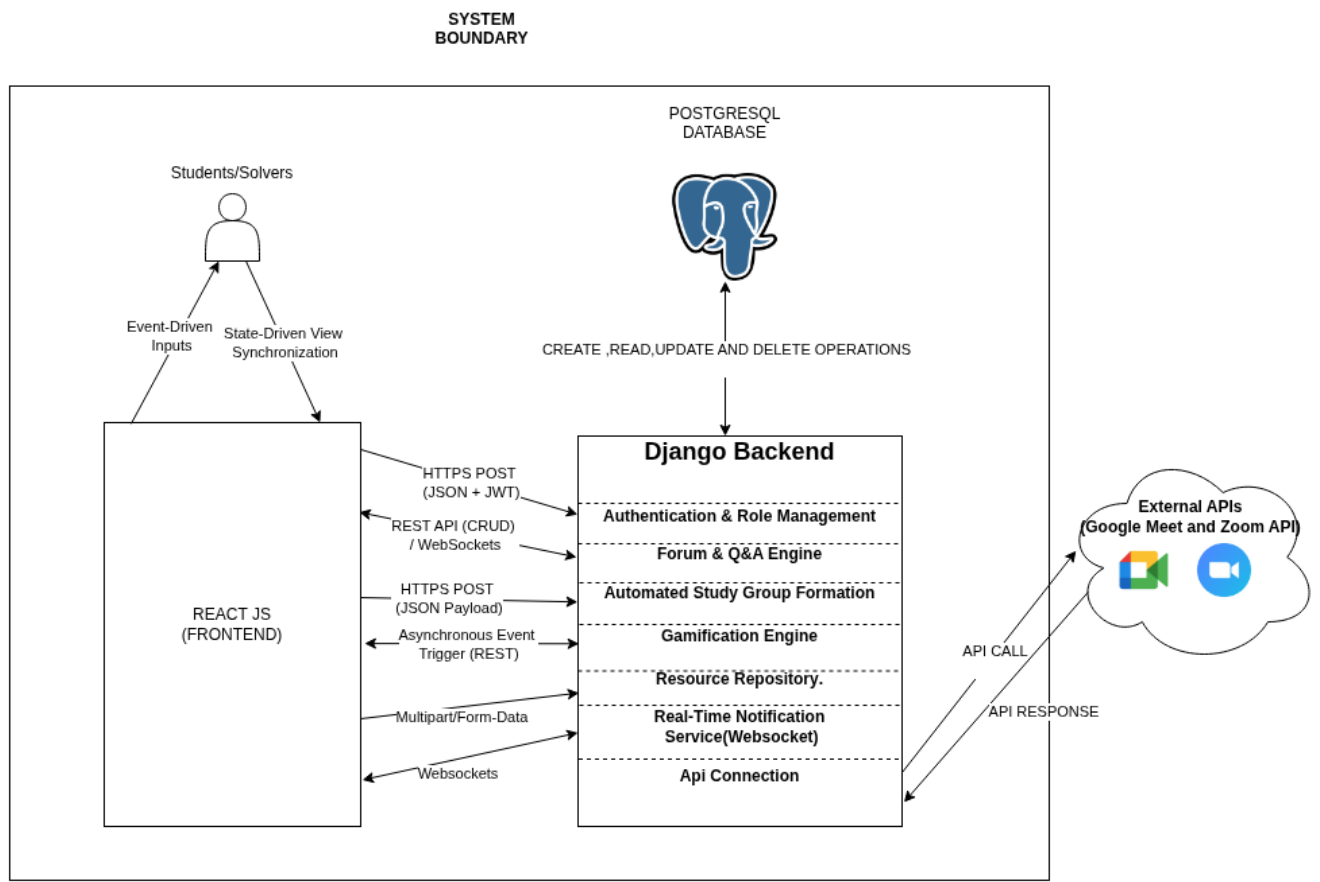


Figure 4, EduConnect Conceptual Framework

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the methodologies and techniques that will be used in the development of the proposed community-driven knowledge-based system for education collaboration. It explains the software development approach selected for the project, the system analysis and design techniques, development tools and expected deliverables. It further describes how the project will be implemented in a structured and systematic manner to ensure the final system will fully identify and solve the identified problem. The selected methodologies, tools and techniques are fully aligned with the objectives and requirements of the proposed EduConnect System.

3.2 System Methodology

The development of the EduConnect system will adopt the Object-Oriented Analysis and Design (OOAD) approach together with the Agile scrum software development methodology.

OOAD is appropriate for EduConnect system since the system has multiple entities, students, academic resources, solvers, meetings and administrators, interacting with each other. The OOAD approach enables the system to be modelled using multiple objects each with their own functionalities and specific attributes therefore improving the reusability, scalability, maintainability and modularity of the software components.

EduConnect will use Agile Scrum methodology because of its support towards iterative development, continuous testing, flexibility and regular feedback from users. Since EduConnect will have real-time communication, collaborative learning and automated meeting generation, the Agile Scrum will provide the flexibility required to continuously improve the system during development.

The development of the proposed EduConnect system will be divided into sprints (Short Iterative cycles). Each sprint will have its own end goal as it will implement specific modules of the system such as authentication, problem-solving forums, study group management, notifications and resource sharing. After every individual sprint, testing and evaluation will be conducted to ensure that the developed functionality meets user requirements.

AGILE METHODOLOGY

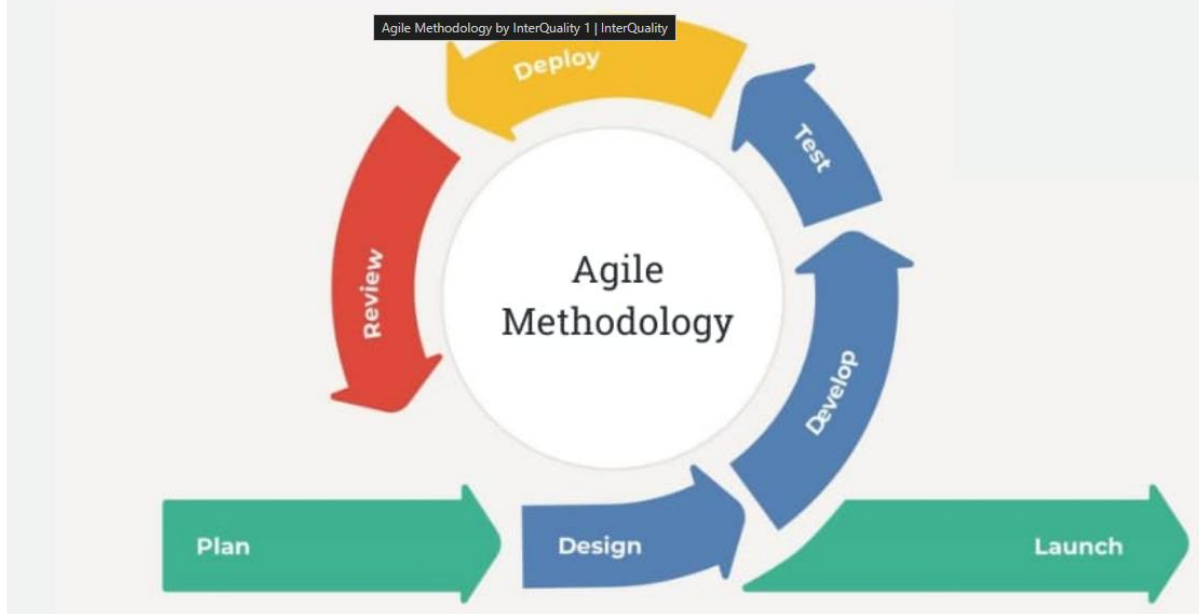


Figure 5, Methodology Diagram

The phase to be followed are as follows:

3.2.1 Requirements Gathering and Analysis

This phase will involve collecting and critically analysing the requirements of the EduConnect system. Through interviews, questionnaires and observation information will be collected to identify the challenges experienced by the students in their collaborative learning and peer-to-peer system. The proposed EduConnect will include a secure user experience by implementing user authentication and registration. To accommodate academic interaction a student will be able to post questions, submit answers and create and manage study groups. The enhancement of engagement will include resource and real-time recommendation and sharing features, supplemented by automated meeting generation. Gamification will be applied to propel consistent participation. Both functional and non-functional requirements will be documented to guide system development

3.2.2 System Design

This phase involves understanding and creating the structure and architecture of the EduConnect platform based on the collected requirements. The following activities will be conducted during this phase: Design Use case diagrams, sequence diagrams, class diagrams, activity diagrams, entity relationship diagrams, user interface wireframes and system architecture. This phase will provide a blueprint that guides the implementation process.

3.2.3 System Development

This is the phase that will involve the coding and implementation of the EduConnect platform using the selected programming languages, frameworks and tools. For the development of the EduConnect system the following will be implemented :Fronted development using React.js, Backend development using Django, Database implementation using PostgreSQL, API integration for Google Meet and Zoom, Real-time communication implementation using WebSockets, Authentication and authorization using JWT and Development of gamification and notification features. The development process will be carried out using sprints to ensure the system is being tested and improved continuously.

3.2.4 System Testing

Testing will be conducted throughout the development lifecycle to ensure the EduConnect system functions correctly and satisfies user requirements. The testing techniques to be implemented are as follows Unit Testing, Integration Testing, System Testing and User acceptance Testing. Through thorough testing we will be able to identify system errors, improve reliability and enhance system security and performance.

3.2.5 Deployment and Maintenance

After thorough and successful testing, EduConnect System will be deployed for use within the academic community. Deployment activities will include the following Server configuration, Database deployment, System installation and configuration and API deployment. Maintenance activities will include Bug fixing, Security updates, Performance

optimization, System upgrades and User support and monitoring This phase will ensure that the system remains secure, reliable and efficient after implementation.

3.2.6 Justification of Methodology

The Agile Scrum methodology is suitable for the EduConnect project since it supports flexibility, continuous improvement and iterative development. The methodology allows for very frequent testing and incorporation of user feedback throughout the development process. Since the EduConnect system contains interactive and dynamic features such as real-time communication, Agile scrum ensures the features will be tested and improved during production.

The OOAD complements Agile Scrum by enabling the system to be designed using reusable and modular objects. This improves maintainability, scalability and efficiency in the system.

3.3 System Analysis

System analysis details the functional and operational requirements of the EduConnect platform. Analysis diagrams will be used to model the system requirements and represent system processes.

3.3.1 Use Case Diagram

A use case diagram is a representation of the various use cases. The use case diagram will illustrate the interactions between users and the EduConnect system. It will represent the functionalities in the EduConnect system such as User registration, posting academic questions, joining study groups, scheduling meetings and managing resources. The diagram will aid in defining the functional scope of the system.

3.3.2 Sequence diagram

Sequence diagrams show the order of interactions between users and system components during different system processes. On the EduConnect platform the sequence diagram will

represent communication flow within the system. From posting academic problems to the related answer submission.

3.3.3 Entity Relationship Diagram

The Entity relationship diagram represents the relationship between the database entities, students, solvers, questions, answers. It will help design an efficient and normalized database structure.

3.3.4 Class Diagram

A class diagram provides a static view of an object-oriented system, showcasing classes, methods, attributes, and relationships among objects (Singh, 2024). The diagram will define system classes, object attributes, methods, functions and relationships between objects.

3.3.5 Activity Diagram

The activity diagram helps visualize workflows, processes or activities within a system. The workflows will include user registration, question posting, group creation and resource allocation.

3.3.6 Data Flow Diagram

The Data Flow Diagram illustrates how information moves within a system. It will illustrate data sources, input and output processes, data storage points and system processing activities. The diagram will help improve understanding of system data processing requirements.

3.4 System Design

System design focuses on transforming system requirements into technical specifications that guide implementation. Thus, section describes the models that will be used to develop EduConnect.

3.4.1 Database Schema

This is a schema that describes the logical constraints that apply to the data stored. It defines views, integrity constraints and tables (Lucid Software Inc., 2021). The schema will define the structure of the EduConnect database. The schema will support efficient storage and retrieval of system data such as user profiles, questions, answers, meetings and notifications.

3.4.2 Wireframes

A wireframe is a crucial blueprint that conveys a systems hierarchy, structure and functionality (ResearchGate,2024). The wireframe will include login and registration pages, student's dashboard, question posting interface. The wireframe will help improve the user experience and the systems usability.

3.4.3 System Architecture

System architecture is the overall structure of the system. EduConnect architecture will move from the frontend layer using React.js to the backend using Django and a Postgres database layer. The architecture design will ensure scalability, maintainability and communication among the various components.

3.5 System Development Tools and Technologies

This section will provide the tools and technologies that will be used during the development of EduConnect.

3.5.1 Visual Studio Code

Visual Studio Code will be the primary Integrated Development Environment to be used during the development. This is because it supports various programming languages, provides debugging tools and improves development efficiency.

3.5.2 Django Framework

Django will be used for backend development as it supports web application development, has strong security features and RESTful API development (Django Software Foundation,2025).

3.5.3 PostgreSQL

PostgreSQL will be the database management tool as it supports relational database management, complex queries and data integrity (PostgreSQL Global Development Group, 2025).

3.5.4 React.js

React.js will be used for frontend development as it supports responsive and interactive user interfaces. The framework supports reusable components and dynamic rendering, making it suitable for modern web applications (React Documentation, 2025).

3.5.5 GitHub

GitHub will be used for version control and source code management. GitHub supports collaboration, source code tracking, backup and project organization (Lonelier & McCullough, 2022).

3.6 Deliverables

This project is expected to produce several tangible deliverables that demonstrate successful completion of the system development process.

3.6.1 Project Proposal

This deliverable will contain the project background, project statement, objectives, literature review and methodology.

3.6.2 System Analysis and Design Documents

This deliverable includes all the diagrams, the database schema, wireframes and system architecture diagrams.

3.6.3 Functional EduConnect System

The project will deliver a fully functional EduConnect platform which will have integrated features such as user registration, resource sharing, peer to peer problem solving, notifications, gamification, and administrator dashboard.

3.6.4 Security and Authentication Features

The system will include secure authentication, password encryption and secure session management.

3.6.5 Testing Documentation

This deliverable will include test case, test and bug reports and user acceptance testing reports.

3.6.6 Final Documentation

The final documentation will include the user manual, technical documentation, installation and maintenance guide.

These documents will enable future use, maintainability and improvements to EduConnect.

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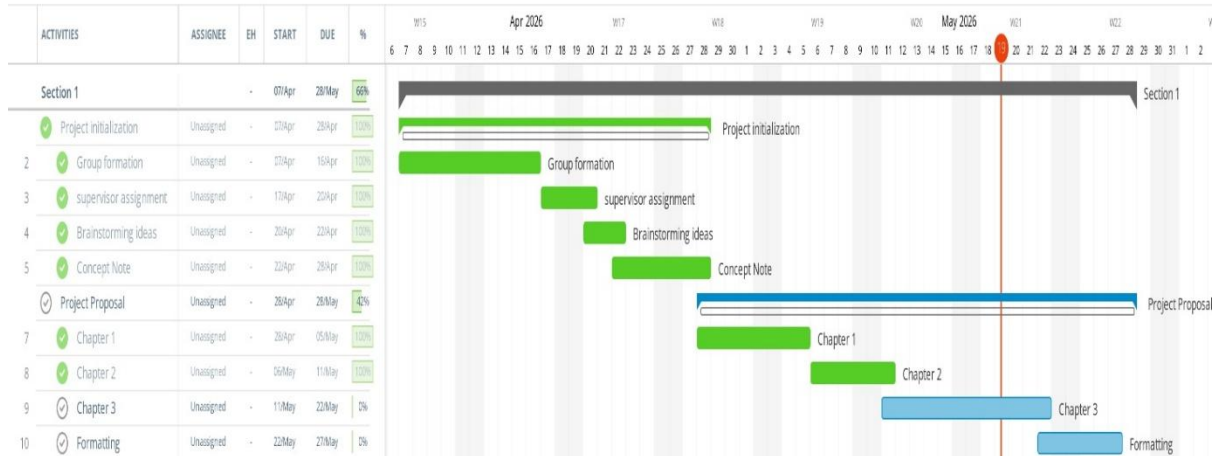
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Appendix

Appendix 1: Gantt Chart

EduConnect

Read-only view, generated on 19 May 2025



Appendix 2: Similarity Report

Include the proposal marking template (uploaded on E-learning) at the very end of the proposal (not as an appendix).